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Atomistic insights into fluoride ion conduction in $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ from ^{19}F PFG-NMR studies

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Abstract

Currently, a variety of solid F^- conductors are being considered as potential electrolytes for the development of all-solid-state fluoride-ion batteries. Among those, PbSnF_4 type solid F^- conductors are promising due to their high ionic conductivity at ambient temperatures. Here we have developed a PbSnF_4 -type solid electrolyte $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ ($x = 0.54$) (BSF), which exhibits high ionic conductivity of approximately $9 \times 10^{-3} \text{ Scm}^{-1}$ at 298 K. In this study, we investigated the origin of its high conductivity and the underlying ion conduction mechanism using nuclear magnetic resonance (NMR) spectroscopy. Using ^{19}F static and magic angle spinning (MAS)-NMR analyses it is revealed that the fluoride ions residing in Sn-rich environment of BSF are highly mobile, and they primarily contribute to the ionic conductivity. Also, the diffusion coefficient (D_{NMR}) measured by pulsed field gradient (PFG)-NMR technique is estimated to be $6 \times 10^{-12} \text{ m}^2\text{s}^{-1}$ at 298 K which is higher than the diffusion coefficient of some well-known lithium ion conducting solid electrolytes. These findings indicate that by using NMR spectroscopic techniques atomistic insights into the ion conduction mechanism and the dynamics could be evaluated successfully in solid F^- conductors.

Keywords

All-solid-state fluoride battery, Solid electrolyte, PbSnF_4 -type fluoride conductor, F diffusion coefficient, PFG-NMR

Introduction

All-solid-state batteries based on fluoride shuttles are promising candidates for commercial rechargeable batteries with higher energy densities compared to those of the currently available battery systems. The fluoride shuttle battery (FSB) functions through the reduction and oxidation of a metal as the anode and metal fluoride as the cathode, releasing electrons per metal atom. An energy density of up to 5000 Wh/L can be achieved by a suitable selection of anode and cathode materials, which is greater than that of any other battery system under consideration [1]. The first prototype FSB was proposed by Schoonman in 1976 [2], and FSB research attracted attention following the work by Reddy et al. reported in 2011 [3]. Various fluoride-ion-conducting solid electrolytes with high ionic conductivities and good chemical stabilities have been considered for FSB development [4-6]. To date, most of the developed solid-state FSBs have operated at 423 K owing to the lower ionic conductivity of the solid electrolyte at ambient temperature [3, 7-9]. Accordingly, further development of fluoride-ion-conducting solid electrolytes that can operate at lower temperatures is a critical requirement for realizing FSBs operating at ambient temperatures.

Several fluoride-ion-conducting solid electrolytes have been developed, which can be classified based on their crystal structure into fluorite-type, tysonite-type, perovskite-type, and PbSnF_4 -type materials [10, 11]. Among these, PbSnF_4 materials have been reported to show high ionic conductivity ($\sim 2.5 \times 10^{-2} \text{ Scm}^{-1}$) at room temperature [12-15]. Along with PbSnF_4 , BaSnF_4 -based compounds have been reported

to be excellent ion conductors at room temperature ($\sim 1 \times 10^{-4} \text{ Scm}^{-1}$) [16, 17]. Different approaches have been used to increase the ionic conductivity of BaSnF_4 , such as varying the Ba-Sn ratio in $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ and modifications of synthesis procedures [17-19]. Further doping of the Ba site in BaSnF_4 with isovalent and aliovalent atoms was considered to increase the ionic conductivity of $\text{Ba}_{1-x}\text{M}_x\text{SnF}_4$ ($\text{M} = \text{Mn}, \text{Nd}, \text{Ce}$) [20-22]. Recently, Mori et al. have successfully achieved a high fluoride ion conductivity of $4 \times 10^{-3} \text{ Scm}^{-1}$ at room temperature by varying the Sn fraction in $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ from 0.5 to 0.53 [23]. The high ionic conductivity of $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ and its increased chemical stability are advantageous for its use as an electrolyte in FSBs [24].

Ionic diffusivity in solid electrolyte plays a key role in determining the battery operational performance and thus information about ionic diffusivity is necessary for selection of solid electrolytes in a battery system. Various experimental techniques can be used for the measurement of fluoride ion (F^-) diffusion coefficients in solids [25-34] and among those, pulsed-field gradient (PFG) NMR has been demonstrated to be an efficient technique for the evaluation of diffusion coefficients in solid electrolytes [35-39]. To date, diffusion coefficients of fluoride ion-conducting solid electrolytes evaluated using field gradient (FG) - NMR have been reported in only a few studies. Most fluoride ion diffusion coefficient measurements are performed at temperatures above 573 K [25, 26, 28-32], primarily due to the low ionic conductivity of these materials at ambient temperatures (See Table 1). As stated earlier, among the fluoride ion conducting solid electrolytes, PbSnF_4 -type materials exhibit high ionic conductivity ($\sim 10^{-3} \text{ Scm}^{-1}$) at ambient temperatures [10, 11]. However, even for PbSnF_4 , the diffusion coefficient measured by PFG-NMR is reported at high temperatures only ($3 \times 10^{-10} \text{ m}^2\text{s}^{-1}$ at 573 K) [40]. To the best of our knowledge, diffusion coefficient measurements at room

temperature for fluoride-ion-conducting solid electrolytes have been rare, which is related to the very short spin-spin relaxation time (T_2), which is a material-dependent physical property.

Table 1: Review of F⁻ diffusion coefficients in fluoride-ion-conducting solids

Material	Sample Nature	Technique	D (m ² s ⁻¹)	Temperature	Ref
LaF ₃	Single crystal	PFG-NMR	1×10^{-13}	848 K	[26]
Na ₃ AlF ₆	Molten Salt	PFG NMR	5.3×10^{-9}	1293 K	[27]
NaF	Polycrystalline	PFG NMR	9.4×10^{-9}	1275 K	[27]
LaF ₃	Nanocrystalline	SFG-NMR*	2×10^{-12}	623 K	[28]
LaF ₃	Nanocrystalline	SFG-NMR	5.5×10^{-13}	650 K	[29]
La _{0.97} Sr _{0.03} F _{2.97}	Nanocrystalline	SFG-NMR	6×10^{-11}	700 K	[30]
LaF ₃	Nanocrystalline	SFG-NMR	2×10^{-11}	650 K	[31]
La _{0.97} Sm _{0.03} F _{2.97}	Nanocrystalline	SFG-NMR	3.3×10^{-10}	700 K	[32]
PbF ₂	Polycrystalline	PFG-NMR	2×10^{-11}	553 K	[25]
PbSnF ₄	Polycrystalline	PFG-NMR	3×10^{-10}	573 K	[40]

*SFG - static field gradient

In addition, the atomistic details and the dynamic process governing high fluoride ion conduction in Ba_{1-x}Sn_xF₂, remain to be fully resolved. Previously, the fluoride ion conduction mechanism in Ba_{1-x}Sn_xF₂ (x = 0.5) was evaluated by Chaudari et al. using temperature dependent NMR spectral analysis and they have reported that the fluoride ions residing in the Ba-rich environment was immobile at ambient temperatures.

Moreover from 2D exchange NMR analysis the mobility of the fluoride ions in the Ba-rich environment is reported to be nearly 10 orders of magnitude lower than the fluoride ions in the Sn-rich environment ($\nu_{F(Sn)} \gg \nu_{F(Ba)}$) indicating that the fluoride ions in the Sn-rich environment to be the major contributor to the conductivity [41]. Apart from this using neutron diffraction (thermal factor) analysis on $Ba_{1-x}Sn_xF_2$ ($x = 0.53$), Mori et al., have identified that the fluoride ions residing in the Sn-rich environment (Ba-Sn layer) are more mobile and based on this they have proposed that the ion conduction in $Ba_{1-x}Sn_xF_2$ ($x = 0.53$) to take place through the zig-zag network motion of the fluoride ions in the Ba-Sn layers [23]. However, there are no reports that have directly investigated the ion dynamics and conduction mechanism in $Ba_{1-x}Sn_xF_2$. As mentioned earlier, the diffusion coefficient measurement by PFG-NMR is highly dependent on the spin-lattice (T_1) and spin-spin (T_2) relaxation time of the materials. Remarkably, the T_1 and T_2 relaxation time for $Ba_{1-x}Sn_xF_2$ ($x = 0.54$) (BSF) is long enough to measure the diffusion coefficient in BSF using ^{19}F PFG-NMR. By combining the PFG-NMR and ionic conductivity measurement results, we aim to clarify the reason for the high conductivity and elucidate the underlying ion conduction mechanism in $Ba_{1-x}Sn_xF_2$.

Experimental methods

A $Ba_{1-x}Sn_xF_2$ ($x = 0.54$) (BSF) solid electrolyte was prepared by mechanical milling. The procedure for sample preparation was described in detail in our previous report [23]. Samples of the same batch were used for conductivity and NMR measurements. AC impedance measurements were conducted using Keysight E4990A impedance analyzer in the frequency range between 100 Hz and 100 MHz with the ac voltage set at 20 mV. A sample pellet with 91% relative density is prepared and platinum

electrodes were sputtered on both sides. Then the sample is loaded to Qualtec ZF003 impedance cell in an argon-filled glove box and taken out for measurements. The impedance measurement was carried out in the temperature range 233 – 333 K.

The prepared polycrystalline BSF solid electrolyte sample was transferred to ϕ -5 mm NMR glass tubes (Shigemi, Japan) under Ar atmosphere inside a glove box. The measurements of the ^{19}F NMR spectra and diffusion coefficients were performed using a Bruker Avance 400 NMR spectrometer (Bruker, Germany) equipped with a Bruker Diff60 gradient probe operating at a resonance frequency of 376.5 MHz. ^{19}F NMR spectra, spin-lattice (T_1) and spin-spin (T_2) relaxation times were measured using single-pulse, inversion-recovery and Carr-Purcell-Meiboom-Gill (cpmg) pulse sequences, respectively. A fluorobenzene ($\text{C}_6\text{H}_5\text{F}$) solution was used as an external standard for referencing the ^{19}F chemical shift ($\delta_{\text{iso}} = -113.15$ ppm). NMR spectra were acquired with a delay time equivalent to five times the T_1 relaxation time. The magic-angle spinning (MAS) NMR spectrum was observed by using a JEOL ECA600 NMR spectrometer operating at 14 T. The resonance frequency for ^{19}F was 564.7 MHz. A MAS probe (Agilent Technologies Inc.) for a 1.6 mm rotor was used with the MAS spinning frequency (ν_{MAS}) of 35 kHz. The ^{19}F chemical shifts were calibrated in ppm relative to CCl_3F adopting the ^{19}F chemical shift for neat C_6F_6 (-163 ppm) as an external reference.

^{19}F diffusion coefficient measurements were performed using the pulsed gradient stimulated echo (PGSTE) method in the temperature range of 298 - 333 K. The pulse-field gradient time (τ) was fixed at 1 ms, and the gradient strength (g) was varied from 37.4 to 600 G/cm (16 steps). The diffusion coefficient was measured over a range of diffusion times ($\Delta = 20 - 50$ ms). For $\Delta > 50$ ms, the signal-to-noise ratio became too low to obtain reliable data, which restricted the measurements to shorter diffusion times.

For the analysis of the diffusion coefficient by the PGSTE method, the Stejskal-Tanner equation [42] was used:

$$I = I_0 \exp(-bD_{NMR}) \quad (1)$$

where I is the signal intensity, I_0 is the initial signal intensity, D_{NMR} is the diffusion coefficient (m^2s^{-1}), and the factor b is given by:

$$b = \gamma^2 g^2 \delta^2 \left(\Delta - \frac{\delta}{3} \right) \quad (2)$$

where γ is the gyromagnetic ratio (Hz/G), g is the gradient strength (G/cm), δ is the gradient pulse duration (s) and Δ is the diffusion time (s).

Results and Discussion

Ionic Conductivity

The ionic conductivity of BSF was investigated using ac impedance measurements over a wide temperature range (233 ~ 333 K). The impedance profile (Nyquist plot) of BSF at different temperatures is shown in Fig 1a. The Nyquist plot at low temperatures (<273 K) exhibits a distorted semicircle in the high frequency region and a spike in the low frequency region (Fig. 1b), which corresponds to contributions from the bulk/grain boundary and electrode, respectively. The bulk and grain boundary resistances were determined through equivalent circuit modeling of the experimental data (see Supplementary Fig. S1), which allowed us to distinguish their respective contributions based on the associated capacitance values (bulk $\sim 10^{-12}$ F, grain boundary $\sim 10^{-9}$ F). At temperatures > 283 K, the impedance which includes bulk and grain boundary resistances, was determined from the intercept of the electrode spike on the Z' -axis. The calculated bulk and the total ionic conductivities within the temperature range measured are presented in Fig. 1c. The total ionic conductivity of BSF at 298 K was found

to be $9.2 \times 10^{-3} \text{ Scm}^{-1}$. Moreover, extrapolation of the bulk ionic conductivity data toward higher temperatures ($> 283 \text{ K}$) indicates that the bulk ionic conductivity at room temperature is 16 mScm^{-1} (Fig. 1c) which is close to PbSnF_4 (See Supplementary Fig. S2) [12]. An Arrhenius plot of ionic conductivity for BSF is shown in Supplementary Fig. S3, where a thermally activated behavior is observed and which can be described by

$$\sigma T = \sigma_0 \exp\left(\frac{E_a^g}{k_B T}\right) \quad (3)$$

here σ_0 is the pre-exponential factor, E_a^g is the activation energy for ionic conductivity, and k_B is the Boltzmann constant. The activation energies for the bulk and total ionic conductivities evaluated using equation 3 were both equal to 0.21 eV . The low activation energy confirms the fast ionic conduction in BSF solid electrolyte. Also, comparison of the ionic conductivity of various $\text{Ba}_{1-x-y}\text{M}_x\text{Sn}_y\text{F}_2$ solid solutions prepared by different techniques (Fig. 1d) shows that ionic conductivity increases substantially with increasing Sn content.

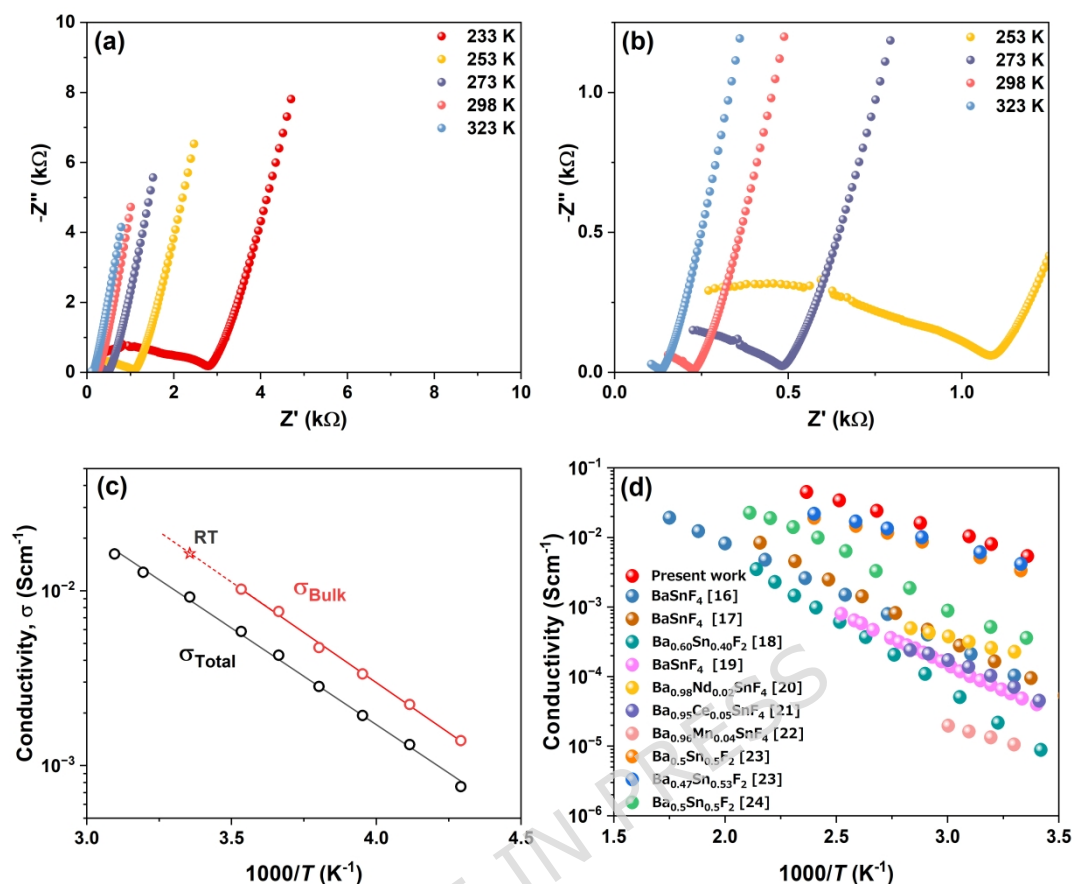


Fig. 1. Fluoride ion conductivity of BSF (a) Nyquist plots of BSF at different temperatures. (b) Magnified Nyquist plots of BSF in the high frequency region. (c) Arrhenius plot of the bulk and total ionic conductivity of BSF (Error bars are on the order of data point size, and represent standard deviation) and (d) Comparison plot showing the ionic conductivity of BSF with various $\text{Ba}_{1-x}\text{M}_x\text{SnF}_2$ (M = Nd, Ce, Mn) compounds [16-24].

Temperature-dependent ^{19}F NMR

The XRD pattern (see Supplementary Fig. S4) of $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ ($x=0.54$) shows narrow peaks, which could be indexed to the crystalline tetragonal phase. Also, a slight secondary phase is also observed which can be indexed to cubic phase of $\text{Ba}_{0.5}\text{Sn}_{0.5}\text{F}_2$. The structure of tetragonal BaSnF_4 is shown in Fig. 2a where the Ba and Sn atoms are ordered in alternative double layers along $[0\ 0\ 1]$ direction of the unit cell and the fluoride ions reside in two anion sub-lattices, one in the Ba–Ba layers and the other one in the Ba–

Sn layers, respectively [23]. For clarity, we refer to fluoride ions residing in the Ba–Ba and Ba–Sn layers as “Ba-rich” and “Sn-rich,” respectively. The ^{19}F MAS-NMR spectrum of BSF measured at room temperature (Fig. 2b) consists mainly of four peaks at 3, -11, -30, and -41 ppm. The ^{19}F NMR resonances observed at 3 and -41 ppm corresponds to the fluoride ions residing in the Ba-rich and Sn-rich environment respectively, which is consistent with those reported for BaSnF_4 [41, 43]. Further, in the present sample, Sn content is higher (0.54) compared to Ba (0.46). Sn is partially incorporated in the Ba-Ba layer, and the ^{19}F resonance signal observed at -30 ppm is attributed to the fluoride ions residing near the Sn incorporated into the Ba-Ba layer. The peak observed at -11 ppm is attributable to fluoride ions in the cubic phase of BaSnF_4 which is present as a minor secondary phase as confirmed from the XRD data (Supplementary Fig. S4) [23]. Additional details regarding the signal assignment are to be published elsewhere.

The ^{19}F NMR spectrum of BSF measured at room temperature under static conditions is presented in Fig. 2c, where a single resonance peak is observed at ~ -43 ppm. The temperature dependence of the full width at half maxima ($\Delta\nu_{\text{F}}$) of the ^{19}F NMR spectra for BSF is shown in Fig. S3 together with the $\Delta\nu_{\text{F}}$ of PbSnF_4 . From Fig. S3, the $\Delta\nu_{\text{F}}$ for BSF shows a plateau with respect to temperature (extreme narrowing region) similar to $\Delta\nu_{\text{F}}$ of PbSnF_4 , indicating that the fast translational motion of the F^- ions. Also, $\Delta\nu_{\text{F}} \sim 2$ kHz observed at ambient temperature for BSF, indicates that the dipolar interaction between the fluoride ions to be partially averaged owing to the fast fluoride ion motion [36, 39]. Additionally, the ^{19}F spin-lattice relaxation time (T_1) for BSF was measured between 298 and 333 K. The T_1 values are observed to decrease with increasing temperature (Supplementary Table S1) indicating increase in mobility with respect to

temperature. These results confirm that fast translational motion of the F⁻ ions occur even at lower temperatures.

According to Chaudari et.al., for BaSnF₄, the fluoride ion residing in the Ba-rich environment is reported to be strongly confined to its site at ambient temperatures and the mobility of these fluoride ions is nearly 10 orders of magnitude lower than that of the fluoride ions in the Sn-rich environment [41]. Hence, appreciable ¹⁹F-¹⁹F dipolar broadening is envisaged for the three peaks observed under MAS condition (Fig. 2b) at 3, -11, and -30 ppm, leading to the absence of these signals under static condition. In other words, under static condition, only the signal from the fluoride ions residing in the Sn rich environment (-41 ppm under MAS condition) is appreciable under static condition. Further, the total fluoride ion concentration for tetragonal Ba_{1-x}Sn_xF₂ (x=0.54) can be estimated by summing up the peak intensities of the ¹⁹F resonances at (3, -30 and -41 ppm). From the peak-fitting analysis of ¹⁹F MAS-NMR spectra, the relative intensities of the peaks corresponding to fluoride ions in the Ba-Sn layer (-41 ppm) and the Ba-Ba layer (3 and -30 ppm) were determined to be 78% and 22%, respectively, consistent with the expected 3:1 ratio of fluoride ions residing in Sn-rich versus Ba-rich environments (Fig. 2a). Our ¹⁹F MAS-NMR results are in good agreement with neutron diffraction results on a similar composition of Ba_{1-x}Sn_xF₂ (x=0.53), where it has been confirmed that the ratio of the fluoride ions residing in the Ba-Sn and Ba-Ba layers is 3:1 [23].

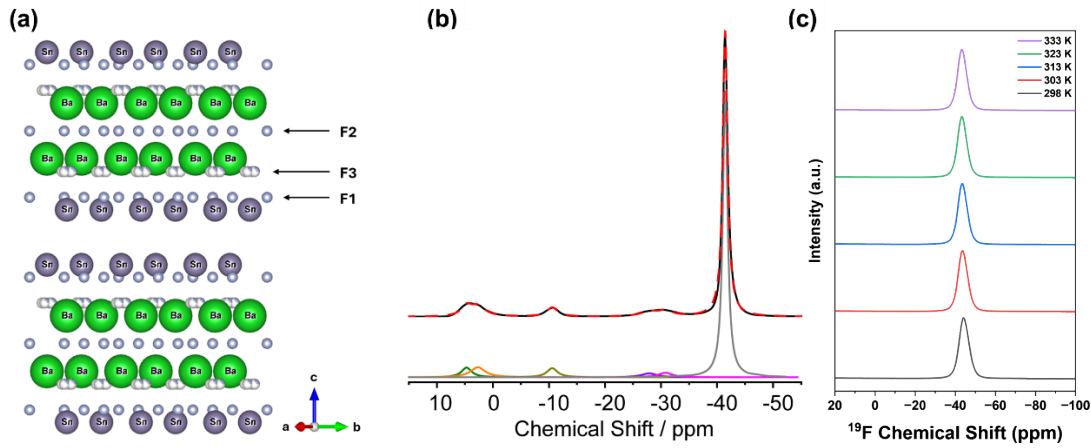


Fig. 2. (a) Structure of tetragonal BaSnF₄ along the 001 direction of the unit cell. (b) ¹⁹F MAS-NMR spectra and decomposition of the observed ¹⁹F MAS-NMR spectrum of Ba_{1-x}Sn_xF₄ (x=0.54) (black line) into six Lorentz lines shown in the bottom. The red dashed line denotes the sum spectrum of the six Lorentz lines. (c) ¹⁹F static NMR spectra of Ba_{1-x}Sn_xF₂ (x=0.54) at different temperatures

Diffusion coefficient measurements using PFG-NMR

The fluoride-ion diffusion coefficient of Ba_{1-x}Sn_xF₂ (x=0.54) was measured by ¹⁹F PFG-NMR spectroscopy at different temperatures using the PGSTE method. Diffusion coefficient measurements were performed by varying the field gradient strength (g) with fixed diffusion time (Δ) and pulse gradient time (τ). Fig. 3a shows the g -dependent PGSTE spectra for BSF acquired at 298 K with a diffusion time $\Delta = 20$ ms and pulse gradient time $\tau = 1$ ms. From Fig. 3a, the PGSTE signal decays with respect to increasing field gradient strength g . Figure 3b, shows the PGSTE signal intensity plotted as a function of $b = \gamma^2 g^2 \delta^2 \left(\Delta - \frac{\delta}{3} \right)$. The fluoride ion diffusion coefficient was estimated by fitting the plot (Fig. 3b) using the Stejskal-Tanner equation (equation 1) and was found to be $6.2 \times 10^{-12} \text{ m}^2\text{s}^{-1}$ at 298 K. This value is greater than the Li-ion diffusion coefficient of well-known garnet-type Li-ion-conducting solid electrolytes at room temperature [36].

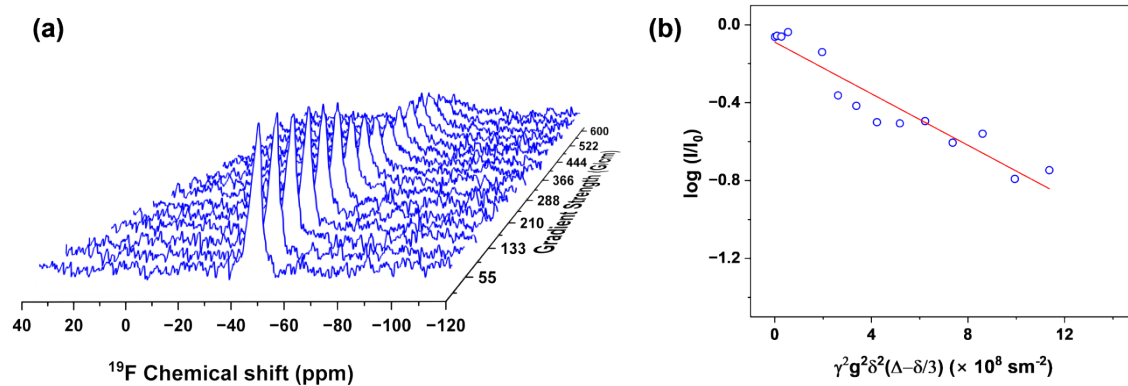


Fig. 3. (a) PGSTE spectra at 298 K as a function of magnetic field gradient strength ‘g’
(b) Stejskal – Tanner attenuation plot

The temperature dependence of the fluoride-ion diffusion coefficient (D_{PFG}) for BSF measured at different temperatures using PFG-NMR (with $\tau = 30 \mu\text{s}$ and $\Delta = 1 \mu\text{s}$) is shown in Fig. 4 (red open squares). The temperature dependence of the diffusion coefficient shows Arrhenius-type thermally activated behavior which can be described by

$$D = D_0 \exp\left(-\frac{E_a^D}{k_B T}\right) \quad (4)$$

where D_0 is the pre-exponential factor, E_a^D is the activation energy of diffusion, k_B is Boltzmann’s constant, and T is the temperature. The activation energy estimated from the temperature-dependent Arrhenius plot (Fig. 4) was found to be 0.23 eV, in excellent agreement with the activation energy estimated from the temperature-dependent ionic conductivity.

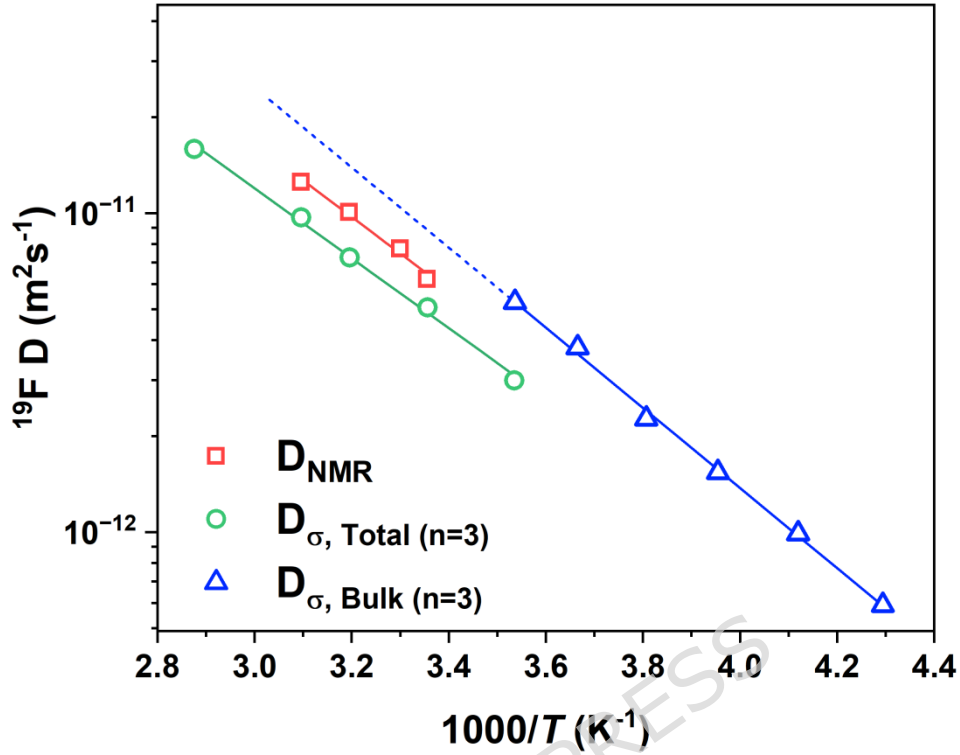


Fig. 4. Temperature dependence of ^{19}F diffusion coefficient for $\text{Ba}_{1-x}\text{Sn}_x\text{F}_4$ ($x=0.54$) (Error bars are on the order of data point size, and represent standard deviation)

Furthermore, to understand the diffusion mechanism of the fluoride ions in BSF, we have calculated the diffusion coefficient from the total ionic conductivity ($D_{\sigma, \text{total}}$) and compared it with the diffusion coefficient evaluated using ^{19}F PFG-NMR. The diffusion coefficient ' D_{σ} ' is related to the ionic conductivity ' σ ' through the Nernst-Einstein relation given by

$$D_{\sigma} = \frac{k_B T}{n(z e)^2} \sigma \quad (5)$$

where n is the number density of ions (cm^{-3}), z is the ion charge (-1 for F^-), e is the elementary charge (1.6021×10^{-19} C), k_B is the Boltzmann's constant (1.38×10^{-23} J/K), T is the absolute temperature (K), and σ is the ionic conductivity (Scm^{-1}).

Here, to calculate the number density of ions, we used the structural parameters for $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ from Ref. [23]. The F^- diffusion coefficient measured by PFG-NMR reflects the mobility of fluoride ions located in the Sn-rich environment, which corresponds to 3 out of the 4 fluoride ions present in the BSF unit cell as confirmed from our MAS-NMR results (Fig. 2c). Based on these observations, we have calculated the number density (n) with 3 fluoride ions (see supplementary note 1) and calculated the diffusion coefficient from ionic conductivity ($D_{\sigma, \text{total}}$) data using equation 5 and shown in Fig. 4 (green open circles), together with the values of the diffusion coefficient (D_{NMR}) measured by PFG-NMR (red squares). Both D_{NMR} and $D_{\sigma, \text{total}}$ show Arrhenius-type thermally activated temperature-dependent behavior. Nevertheless, the diffusion coefficient ($D_{\sigma, \text{total}}$) calculated from the total ionic conductivity is lower than the diffusion coefficient (D_{NMR}) measured using PFG-NMR ($D_{\text{NMR}} \sim 1.6 \times D_{\sigma, \text{total}}$). Moreover, from the ionic conductivity results (Fig. 1c), a contribution from the bulk of the electrolyte to the ionic conductivity is observed at temperatures below room temperature. Therefore, we calculated the diffusion coefficient corresponding to the bulk ionic conductivity ($D_{\sigma, \text{Bulk}}$) using equation 5 (Fig. 4 (blue open triangles)) and compared it with the diffusion coefficient measured by PFG-NMR (D_{NMR}). It is clear from Fig. 4 that the diffusion coefficient calculated from bulk ionic conductivity data ($D_{\sigma, \text{Bulk}}$) to be greater (blue dashed line) than the diffusion coefficient measured by PFG-NMR ($D_{\sigma, \text{Bulk}} > D_{\text{NMR}}$) when extrapolated to higher temperatures. This indicates that the fluoride-ion diffusion coefficient measured using PFG-NMR primarily reflects the contribution from the bulk (grain) of the material. The higher diffusion coefficient value for the bulk ionic conductivity compared with the measured diffusion coefficient indicates that fluoride ion conduction occurs through the correlated motion of the fluoride ions in BSF [36, 37, 39].

As shown in Fig. 2a, fluoride ions reside in 3 sites, one in the Ba-Ba (Ba-rich, F2) layer and the other two in the Ba-Sn (Sn-rich, F1 & F3) layer. Based on our results, we propose a fluoride-ion migration mechanism in which an F1-site fluoride hops to a neighboring F3 site, triggering the F3-site ion to move into an adjacent F1 site. This sequence produces a non-collinear, correlated (-F1-F3-F1-) motion involving multiple fluoride ions, giving rise to the two-dimensional fast F⁻ diffusion in BSF [23].

In addition, to validate that the main contribution to the diffusion coefficient measured by PFG-NMR to be from the bulk of the material, we calculated the root mean square displacement (RMSD) of fluoride ions based on the diffusion time dependent diffusion coefficient values using the following relation:

$$\text{RMSD} = \sqrt{D \Delta t} \quad (7)$$

where D is the diffusion coefficient (D_{NMR}) and Δt is the diffusion time at a given temperature. In this study, the diffusion time (Δt) varied from 20 to 50 ms during the diffusion coefficient measurements using PFG-NMR. These results indicated that the diffusion coefficient decreased with increasing diffusion time (Table 2). The slight decrease in the diffusion coefficient (D_{NMR}) with respect to the diffusion time (Δt) is most likely related to the grain size effect, which restricts the diffusion length to the grain size [36]. The RMSD of the fluoride ions was found to be ~ 0.5 μm within the measured time range [Table 2 and Supplementary Fig. S6]. As shown in Supplementary Figure S7, particles with sizes < 1 μm were observed in the samples used in this study. Therefore, it is concluded that the diffusion of the fluoride ions in the bulk of the material is the main contribution to the diffusion coefficient of the fluoride ions measured by PFG-NMR and ionic conductivity measurements.

Table 2. Root mean square displacement of fluoride ions in $\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ ($x=0.54$).

Diffusion time (Δt), ms	Diffusion coefficient (D_{NMR}), $\times 10^{-11} \text{ m}^2 \text{ s}^{-1}$	Root Mean square displacement (RMSD), (μm)
20	1.17	0.48
30	0.78	0.48
40	0.65	0.51
50	0.54	0.52

In this work, we succeeded in measuring the fluoride ion diffusion coefficient in BSF, a superionic conductor using ^{19}F PFG-NMR technique at room temperature which is a first. Also, we have directly identified that the high conductivity observed for BSF is due to the correlated motion of the fluoride ions residing in the Sn-rich environment in BSF.

Conclusion

$\text{Ba}_{1-x}\text{Sn}_x\text{F}_2$ ($x=0.54$) is found to exhibit very high fluoride ion conductivity of $9.2 \times 10^{-3} \text{ S cm}^{-1}$ at 298 K. The major ion conduction carrier was identified as the fluoride ions in the Sn-rich environment with a diffusion coefficient of $6 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$ at 298 K. In addition, MSD analysis indicated that PFG-NMR predominantly detected the diffusion of fluoride ions in the bulk material. Thus, the findings obtained in this study provide fundamental insight into controlling the ionic conductivity in BSF suitable for application in all solid-state batteries.

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Author contributions

AD: Investigation, performed the static & PFG-NMR experiments, SEM observation, data curation, and writing original draft preparation; **RT:** Investigation and reviewed the manuscript; **AM:** sample preparation, performed the impedance experiments and reviewed the manuscript; **MM:** performed the MAS-NMR experiments, investigation, reviewed and edited the manuscript; **TO:** funding acquisition, supervision, reviewed and edited the manuscript; **JK:** Methodology and reviewed the manuscript. The authors have read and approved the manuscript.

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Data Availability

Data supporting this study’s findings are available from the corresponding author (A.D.) upon reasonable request.

Declarations

Competing Interests

The authors declare no competing interests.

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